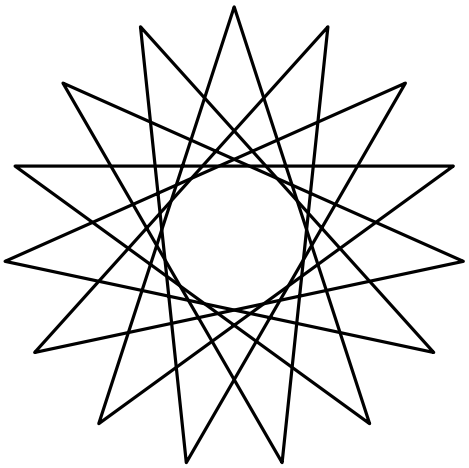
Problem 5: Write a message of at least six words, and put it into code with a setting you keep secret. Give only the code to a partner, who finds the message.

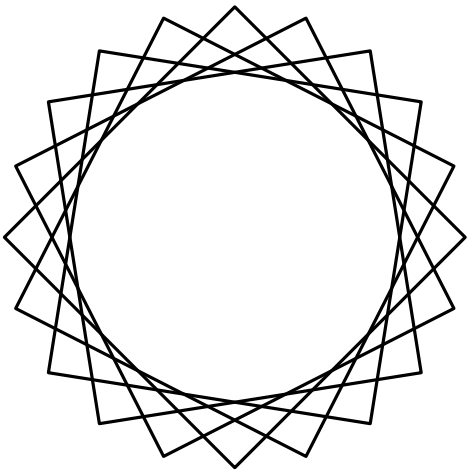
Problem 8: For which rings, from 4 dots up to 20 dots, does every hop smaller than the number of dots draw a line to every dot without starting again? Circle them, and explain why your list is complete.

4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

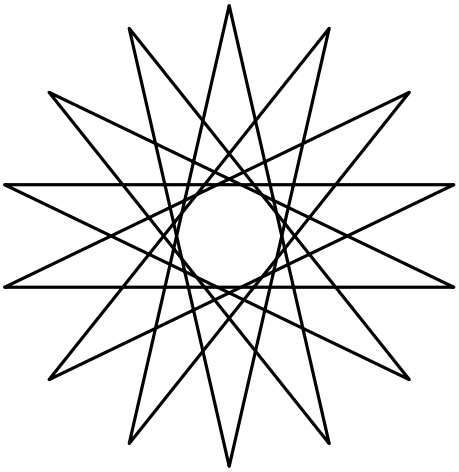
Problem 9: Each of these drawings was made on a ring of dots with one hop, and then the dots were erased. Find the number of dots and a hop for each drawing.



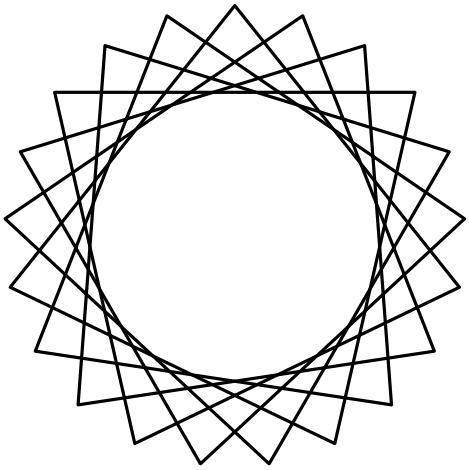
dots _____ hop _____



dots _____ hop _____



dots _____ hop _____

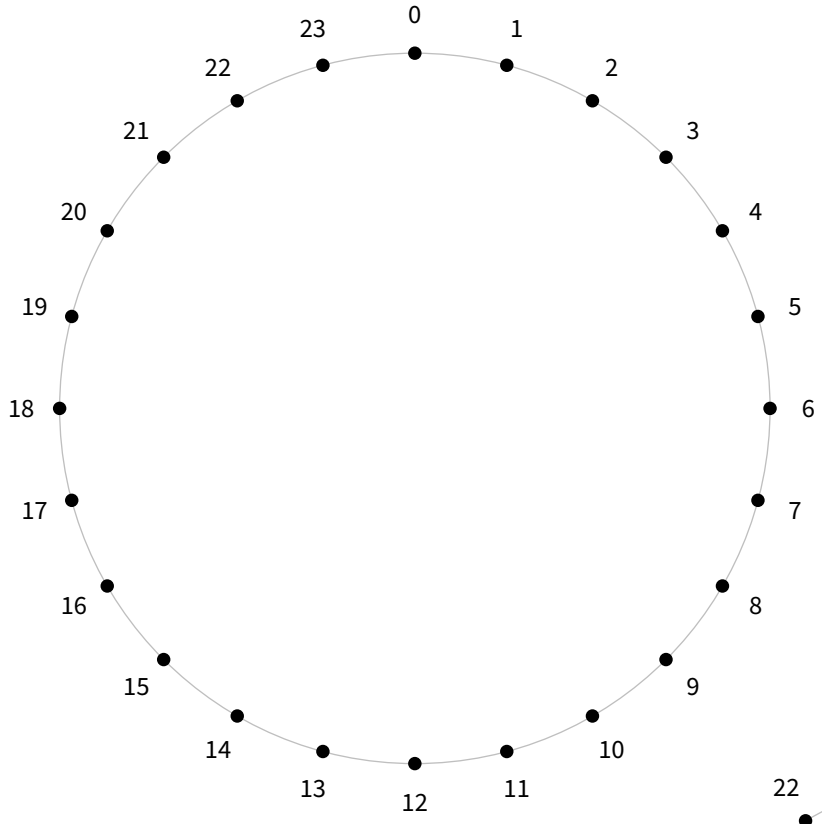


dots _____ hop _____

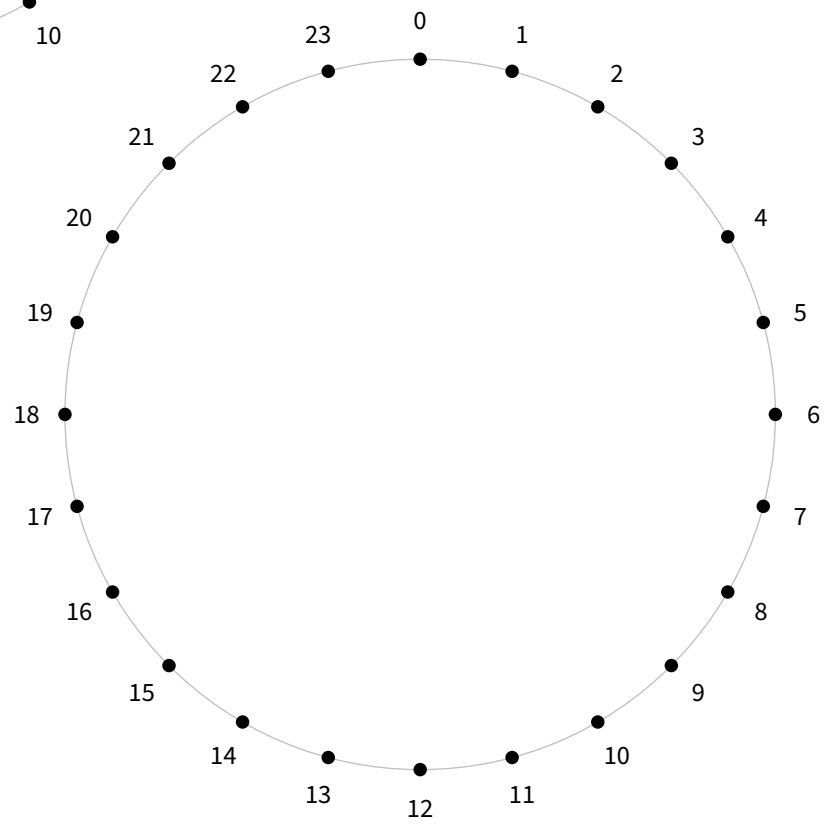
Problem 10: Lee puts each message into code twice, with two different settings, so that it will be harder to crack. Explain whether this makes his messages any harder to crack.

Problem 11: Set the wheel to C. Put A into code, then put the new letter into code, and keep going until you get back to A. Which settings take A through all 26 letters before it comes back? Explain why your list is complete.

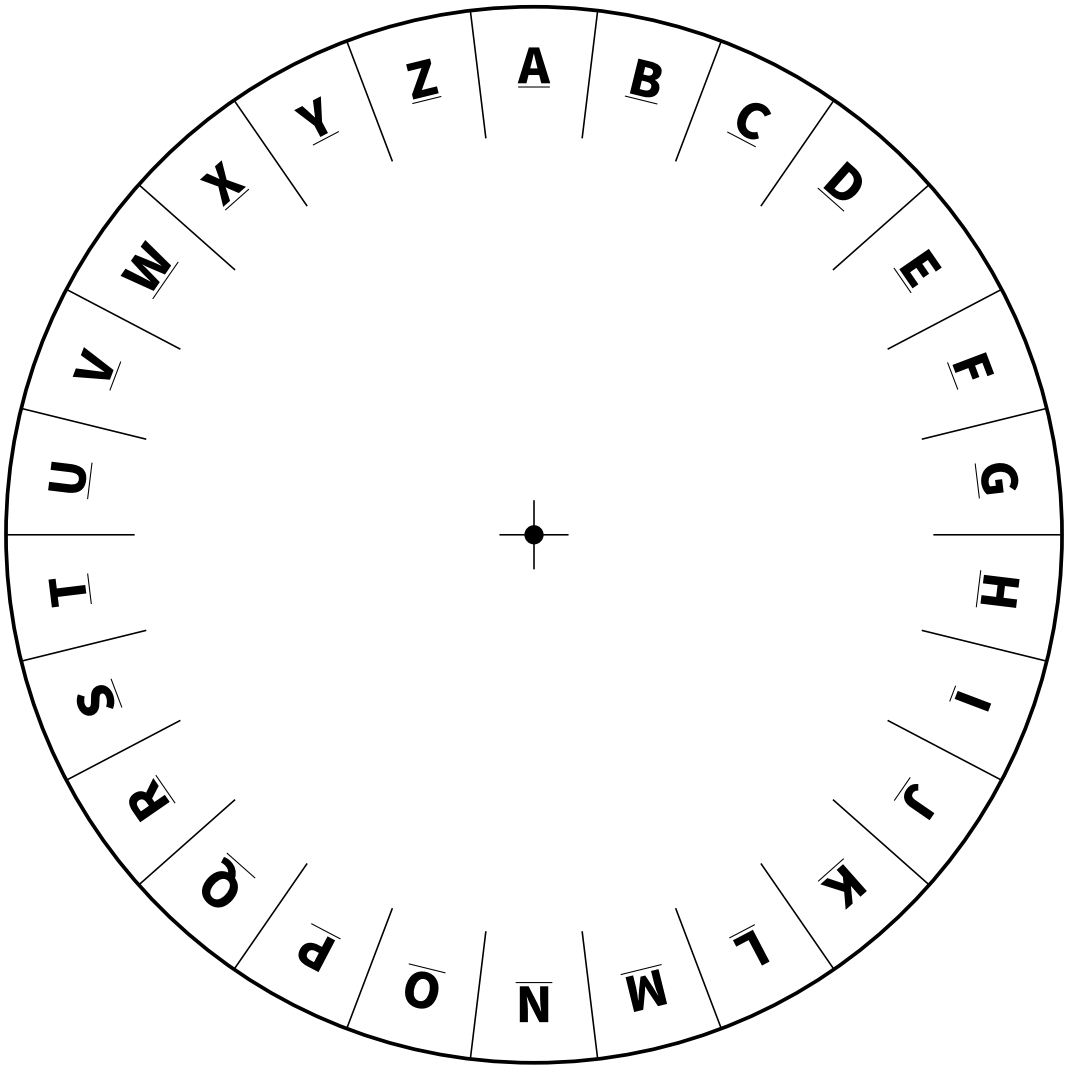
Problem 12: On the ring marked ×2, draw a line from each dot to the dot with twice its number, counting on around the ring past 23 when you need to, so that dot 13 joins dot 2. On the ring marked ×3, draw a line from each dot to the dot with three times its number. A dot that lands on itself gets no line.

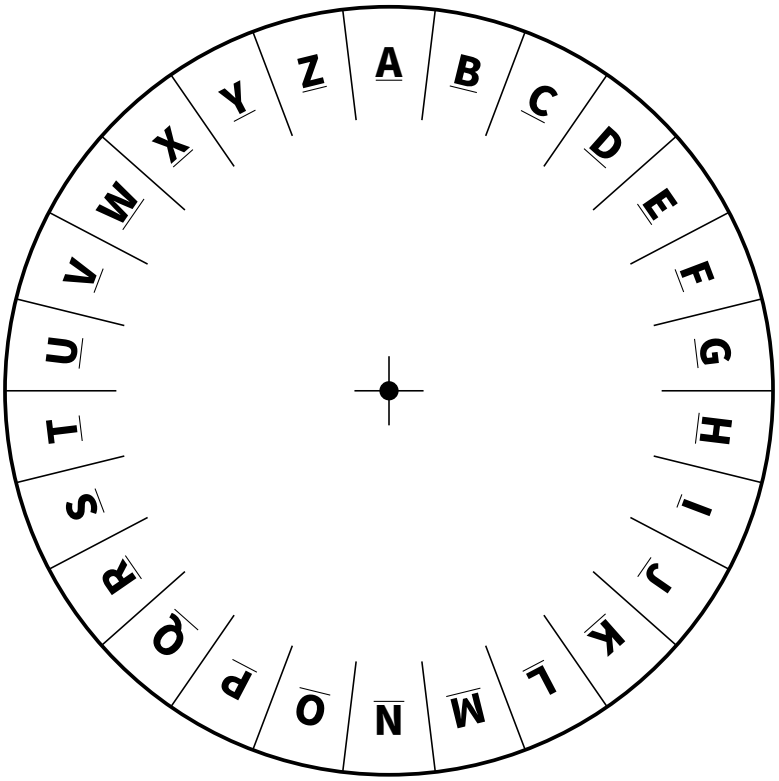


×2



×3


A
B
C
D
E
F
G
H
I
J
K
L
M
N
O
P
Q
R
S
T
U
V
W
X
Y
Z


A
B
C
D
E
F
G
H
I
J
K
L
M
N
O
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Q
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S
T
U
V
W
X
Y
Z